

Lecture 6 - When Complete Randomization Fails

Attrition after Random Assignment

Dr. Xi Chen

Rotterdam School of Management

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Outline

- 1 Attrition and the conventional approaches
- 2 Bounding approach
- 3 Bounding approach in attrition

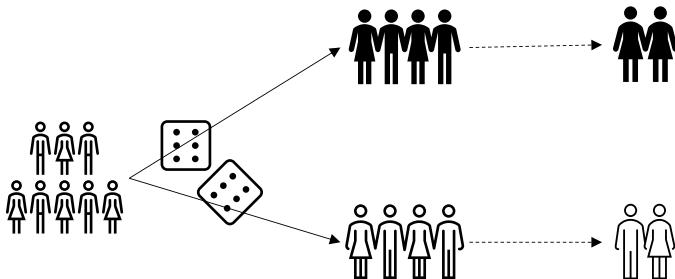
1 Attrition and the conventional approaches

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3 Bounding approach in attrition

Failure to measure outcomes

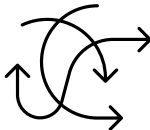
The general idea of attrition: the failure to measure outcomes of some units.



Various reasons of attrition



Refuse to cooperate
(two-stage studies)



Lose track of subjects
(long-term studies)



Blocked by authorities
(field experiments)



Unmeasurable
(e.g., decease)

Attrition: an example

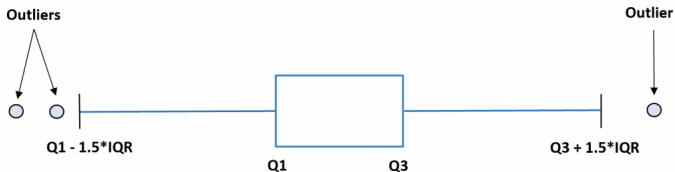
The introduction of e-commerce to countryside in China¹.



Attrition: **local governments denied access to some villages.**

¹Please check Couture et al. (2021) for more details.

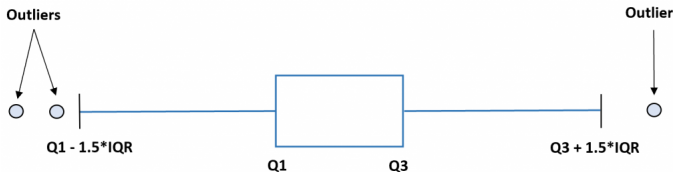
Attrition: the deletion of outliers in experimental data



Should we delete outliers in experimental data?

- Deletion leads to missing outcomes for some units.

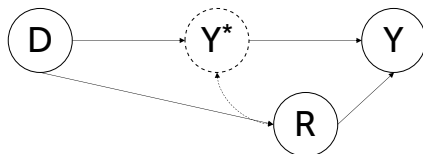
Attrition: the deletion of outliers in experimental data



Should we delete outliers in experimental data?

- Deletion leads to missing outcomes for some units.
- Leading to the failure of randomization, eventually the non-identification of ATE.

A DAG representation of attrition



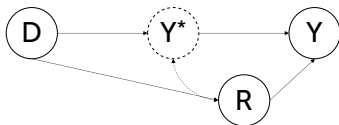
- D is the treatment, and **exogenous due to randomization**.
- Y^* is a latent outcome, partially observed by researchers.
- $R = \{0, 1\}$ is the binary state of attrition, caused by or causing Y^* .
- Y is the observed outcome with $Y = (1 - R) \times Y^*$

Fact (Intention-to-treat)

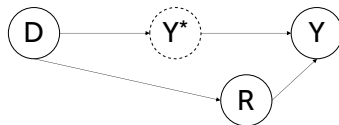
Without any assumption, we can identify the intention-to-treat effects with $P(Y | D)$.

Traditional approaches: Random attrition

Assuming the attrition is random...



(a) DAG of Non-random Attrition



(b) DAG of Random Attrition

Fact (Random Attrition)

Under random attrition, we can identify the average treatment effect with $P(Y^ | D) = P(Y^* | D, R)$.*

Traditional approaches: Random attrition

Technical proof:

$$\begin{aligned}P(Y^* \mid do(D)) &= P(Y^* \mid D, R = 0) P(R = 0) + P(Y^* \mid D, R = 1) P(R = 1) \\&\stackrel{Y^* \perp R \mid D}{=} P(Y^* \mid D) P(R = 0) + P(Y^* \mid D) P(R = 1) \\&= P(Y^* \mid D)\end{aligned}$$

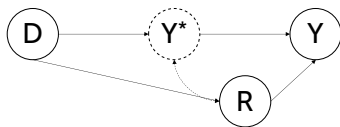
Because the attrition is at random, we can use observed outcome Y to calculate the conditional distribution:

$$P(Y^* \mid D) = P(Y^* \mid D, R = 0) = P(Y \mid D)$$

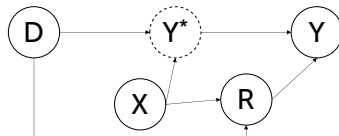
Traditional approaches: imputation

Assumption for Imputation

We observe **non-descendant** variables X that **separate** Y^* and R .



(a) DAG of Non-random Attrition

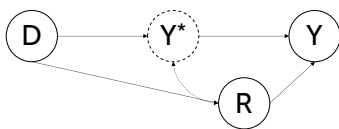


(b) DAG of Imputation Methods

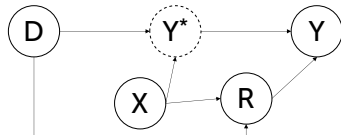
Traditional approaches: imputation

Assumption for Imputation

We observe **non-descendant** variables X that **separate** Y^* and R .



(a) DAG of Non-random Attrition



(b) DAG of Imputation Methods

Fact (Imputation Method)

Under the assumption of the imputation method, we can identify the average treatment effect with
$$P(Y^* \mid D, X) = P(Y^* \mid D, X, R).$$

Traditional approaches: imputation

Technical proof:

$$\begin{aligned}
 P(Y^* \mid do(D), X) &= P(Y^* \mid D, X, R = 0) P(R = 0) + P(Y^* \mid D, X, R = 1) P(R = 1) \\
 &\stackrel{Y^* \perp R \mid X}{=} P(Y^* \mid D, X) P(R = 0) + P(Y^* \mid D, X) P(R = 1) \\
 &= P(Y^* \mid D, X)
 \end{aligned}$$

Because the attrition is at random conditional on X , we can use observed outcome Y to calculate the conditional distribution:

$$P(Y^* \mid D, X) = P(Y^* \mid D, X, R = 0) = P(Y \mid D, X)$$

We can then integrate over X to have the ATE:

$$P(Y \mid D, X) = \sum_{x \in X} P(Y \mid D, X = x)$$

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- 1 Attrition and the conventional approaches
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Relaxing unrealistic assumptions

Many assumptions are non-refutable and unrealistic.

- Unconfoundedness
- Random attrition
- ...

Forego the point identification, make weaker assumptions and increase credibility of analyses.

Strong assumptions identify a point: •

Weak assumptions identify a set: [— — ——]

Some types of bounds: no-data bounds

Assume the potential outcomes Y^0 and Y^1 are bounded $[0, 1]$.

$$-1 \leq Y_i^1 - Y_i^0 \leq 1 \Rightarrow E(Y_i^1 - Y_i^0) \in [-1, 1]$$

In more general terms, if Y^0 and $Y^1 \in [a, b]$

$$E(Y_i^1 - Y_i^0) \in [a - b, b - a]$$

The length of the bound is $2(b - a)$.

Some types of bounds: no-assumption bounds

Assume we collect some data with $\{Y_i, D_i\}$, but random assignment is not ensured².

Can we do better with data?

$$\begin{aligned} E(Y_i^1 - Y_i^0) &= \underbrace{\pi E(Y_i^1 | D_i = 1)}_{\text{Observed}} + \underbrace{(1 - \pi) E(Y_i^1 | D_i = 0)}_{\text{Unobserved}} \\ &\quad - \underbrace{\pi E(Y_i^0 | D_i = 1)}_{\text{Unobserved}} - \underbrace{(1 - \pi) E(Y_i^0 | D_i = 0)}_{\text{Observed}} \end{aligned}$$

Under the bounds on potential outcomes $[a, b]$:

$$\tau^{obs} + (1 - \pi) a - \pi b \leq E(Y_i^1 - Y_i^0) \leq \tau^{obs} + (1 - \pi) b - \pi a$$

The length of the bound is $(b - a)$, half of the no-data bounds.

²Let $\pi = P(D_i = 1)$ in the data collected.

Some types of bounds: monotone treatment response

Assume treatment always benefits people, or $Y_i^1 \geq Y_i^0, \forall i$
This means ITE is nonnegative, so the lower bound of ITE is 0.
Under this assumption, the lower bound of ATE is also 0:

$$E(Y_i^1 - Y_i^0) \geq 0$$

Using this assumption, we may further narrow the no-data and no-assumptions bounds.

Fact

Under monotone treatment response assumption, the no-data bound becomes $E(Y_i^1 - Y_i^0) \in [0, b - a]$.

Some types of bounds: monotone treatment selection

Assume the treatment group potential outcomes are better than control groups, with:

$$\begin{aligned}E(Y_i^1 | D_i = 1) &\geq E(Y_i^1 | D_i = 0) \\E(Y_i^0 | D_i = 1) &\geq E(Y_i^0 | D_i = 0)\end{aligned}$$

Under this assumption, the ATE is bounded from above,

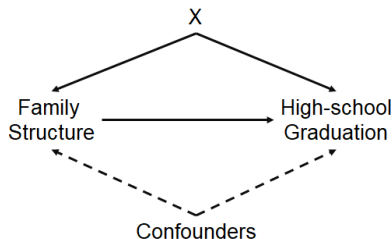
$$E(Y_i^1 - Y_i^0) \leq E(Y_i^1 | D_i = 1) - E(Y_i^0 | D_i = 0)$$

We may use it to further narrow the no-assumption bounds.

Bounding approach in practice

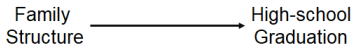
Manski et al. (1992) studied how family structure during adolescence influences the high school graduation rate.

Data from National Longitudinal Survey of Youth: D (if living with both parents), Y (if graduate by 20), X (a set of control variables).

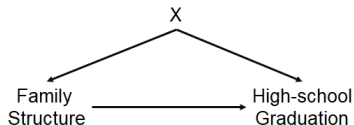


Assumption 1 & 2: No confounders / Selection observables

Variable	Exogenous	Selection on X
Constant	0.47 (5.81)	0.51 (11.61)
<i>Race and ethnicity</i>		
<i>Parental education</i>		
<i>Residence</i>		
Nonintact family at 14	-0.36 (6.75)	-0.38 (2.27)



(a) No confounders

(b) Selection on X

Family structure: no assumption bounds

	Intact Family	Non-Intact Family
Graduated	1	2
Not Graduated	3	4

- **Lower bound:** Flipping the outcome of in Cell 1 and 4.
- **Upper bound:** Flipping the outcome of in Cell 2 and 3.

Estimation results: $[-0.74, +0.20]$.

Family structure: monotone treatment selection

For any value of unobserved confounders, any youth when put in an intact family, are at least as likely to graduate as when not.

$$E(Y_i^1 \mid D_i = 1) \geq E(Y_i^1 \mid D_i = 0)$$

$$E(Y_i^0 \mid D_i = 1) \geq E(Y_i^0 \mid D_i = 0)$$

Estimation results: $[-0.31, -0.02]$.

Family structure: comparing difference approaches

Approach	Estimation
Exogenous	-0.36
Selection on Observables	-0.38
Monotone Treatment Selection	$[-0.31, -0.02]$
No Assumptions Bounds	$[-0.74, +0.20]$

Law of Decreasing Credibility

Stronger conclusions require stronger assumptions that have a lower level of credibility.

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A running example

- We have a data from a randomized experiment.
- The outcome Y_i measured with a scale from 1 to 10 (i.e., $Y_i \in [1, 10]$).
- The experiment has an attrition problem, with NA denoting missing values.

Obs	D_i	Y_i
1	1	10
2	1	8
3	1	NA
4	0	3
5	0	NA
6	0	6

$$Y_i \in [1, 10]$$

A running example

Obs	D_i	Y_i
1	1	10
2	1	8
3	1	NA
4	0	3
5	0	NA
6	0	6

$$Y_i \in [1, 10]$$

- Without additional assumptions, the intention-to-treat is:

$$\text{ITT}_Y = \frac{10+8}{3} - \frac{3+6}{3} = 3.$$

- Under random attrition, the ATE is:

$$\text{ATE}_{\text{random}} = \frac{10+8}{2} - \frac{3+6}{2} = 4.5.$$

Bounding approach: Manski bounds

Manski bounds have no additional assumptions. See Horowitz and Manski (2000).

- Upper bound: substituting NA's in **treatment with max values and control with min values**.
- Lower bound: substituting NA's in **treatment with min values and treatment with max values**.

Obs	D_i	Y_i
1	1	10
2	1	8
3	1	NA
4	0	3
5	0	NA
6	0	6

$$Y_i \in [0, 10]$$

Bounding approach: Manski bound

Obs	D_i	Y_i
1	1	10
2	1	8
3	1	NA
4	0	3
5	0	NA
6	0	6

- Manski bound: $\left[\frac{10+8+1}{3} - \frac{3+10+6}{3}, \frac{10+8+10}{3} - \frac{3+1+6}{3} \right] = [0, 6]$
- $ITT_Y = 3$ and $ATE_{\text{random}} = 4.5$

In practice, use *max* and *min* values of observed outcomes.
Uninformative but wide, as a benchmark for further analysis.

Can we narrow the bound?

Lee bounds Lee (2009)

Assumptions: **monotonicity of treatment selection.**

- Treatment status either increases / decreases attrition for all people.
- The effect of treatment on attrition has the same sign (+ or -).

How to use the condition?

- Trimming samples such that share of observed individuals is equal in the treatment and control.
- Trimming from above or below to obtain lower or upper bound.

Lee bounds: key idea

- Suppose we have a treatment group with 40% of attrition and a control group with 50%.

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- To get the same level of “missing,” we need to trim $x\%$ of people from the treatment.

Lee bounds: key idea

- Suppose we have a treatment group with 40% of attrition and a control group with 50%.
- The treatment decreases the attrition likelihood for 10%.
- To get the same level of “missing,” we need to trim $x\%$ of people from the treatment.
- To calculate $x\%$, we do:

$$\frac{40\%}{1 - x\%} = 50\% \Rightarrow x\% = \frac{50\% - 40\%}{50\%} = 20\%$$

Lee bounds: key idea

Question: **which 20% of the treated units to trim?**

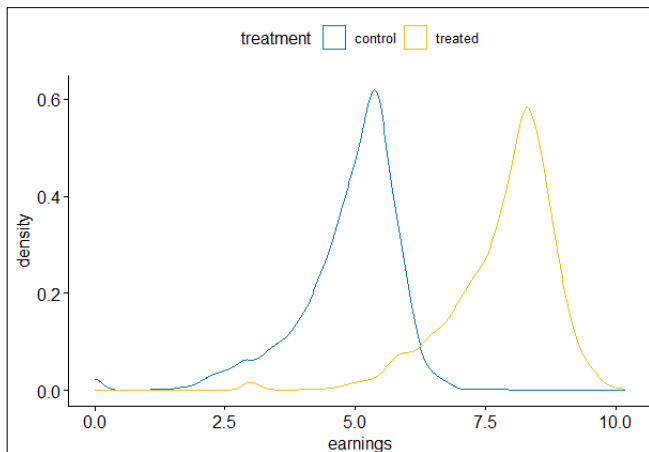
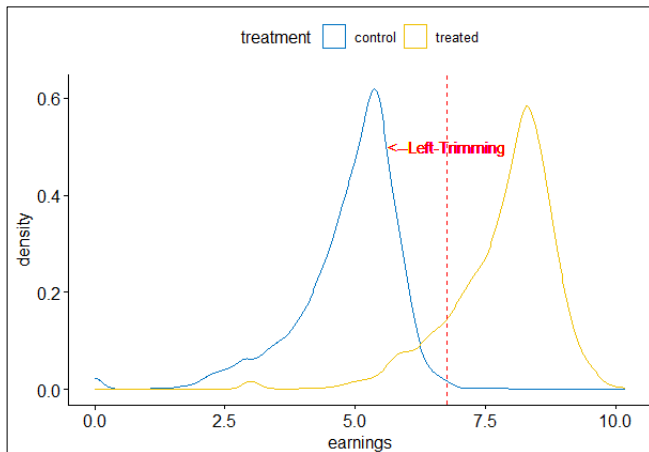


Figure 4: Density Plots of Outcomes for the Treated and Control Group

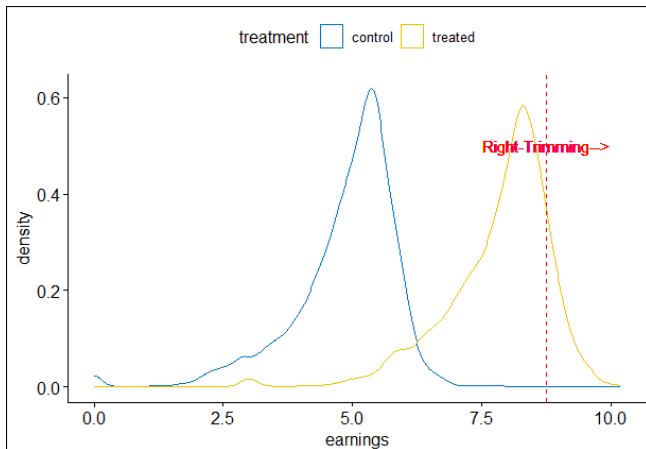
Lee bounds: key idea

To trim the people in the 20% lower quantile to get the upper bound:



Lee bounds: key idea

To trim the people in the 20% upper quantile to get the lower bound:



Lee bounds: procedure

Y the outcomes, D the treatment, and W the non-attrition.

- Step 1: Get the share of individuals with observed Y in the treatment and control, as q_1 and q_0 .
- Step 2: Obtain the attrition difference:

$$q = \begin{cases} \frac{q_1 - q_0}{q_1} & \text{if } q_1 > q_0 \\ \frac{q_0 - q_1}{q_0} & \text{if } q_0 > q_1 \end{cases}$$

- Step 3: Find the q th and $(1 - q)$ th quantile of observed outcomes in the treatment group as Y_q^1 and Y_{1-q}^1 .

Lee bounds: procedure

Y the outcomes, D the treatment, and W the non-attrition.

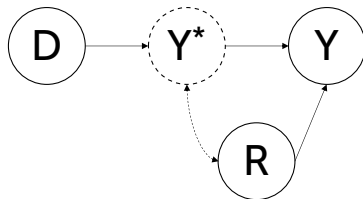
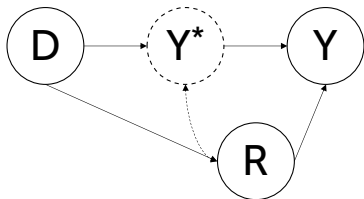
- Step 4: Upper and lower bound:

$$\begin{aligned} \bar{\tau} &= \frac{\underbrace{\sum 1(Y_i \geq Y_q^1) D_i W_i Y_i}_{\text{Non-attriters in treatment with higher outcome}}}{\underbrace{\sum 1(Y_i \geq Y_q^1) D_i W_i}_{\text{Non-attriters in treatment with higher outcome}}} - \underbrace{\frac{\sum (1 - D_i) W_i Y_i}{\sum (1 - D_i) W_i}}_{\text{Non-attriters in control}} \\ \underline{\tau} &= \frac{\underbrace{\sum 1(Y_i \leq Y_{1-q}^1) D_i W_i Y_i}_{\text{Non-attriters in treatment with lower outcome}}}{\underbrace{\sum 1(Y_i \leq Y_{1-q}^1) D_i W_i}_{\text{Non-attriters in treatment with lower outcome}}} - \underbrace{\frac{\sum (1 - D_i) W_i Y_i}{\sum (1 - D_i) W_i}}_{\text{Non-attriters in control}} \end{aligned}$$

Lee bounds: summary

Treatment influences attrition in one direction:

- By comparing the treatment and control, we may “partial out” the effect of treatment on attrition.
- ATE on the non-attriters!



References I

-  Couture, Victor, Benjamin Faber, Yizhen Gu, and Lizhi Liu. 2021. "Connecting the Countryside via E-Commerce: Evidence from China." *American Economic Review: Insights*, 3 (1): 35-50.
-  Horowitz, J. L., & Manski, C. F. (2000). Nonparametric analysis of randomized experiments with missing covariate and outcome data. *Journal of the American Statistical Association*, 95(449), 77-84.
-  Lee, D. S. (2009). Training, wages, and sample selection: Estimating sharp bounds on treatment effects. *The Review of Economic Studies*, 76(3), 1071-1102.

References II

-  Manski, C. F., Sandefur, G. D., McLanahan, S., & Powers, D. (1992). Alternative estimates of the effect of family structure during adolescence on high school graduation. *Journal of the American Statistical Association*, 87(417), 25-37.